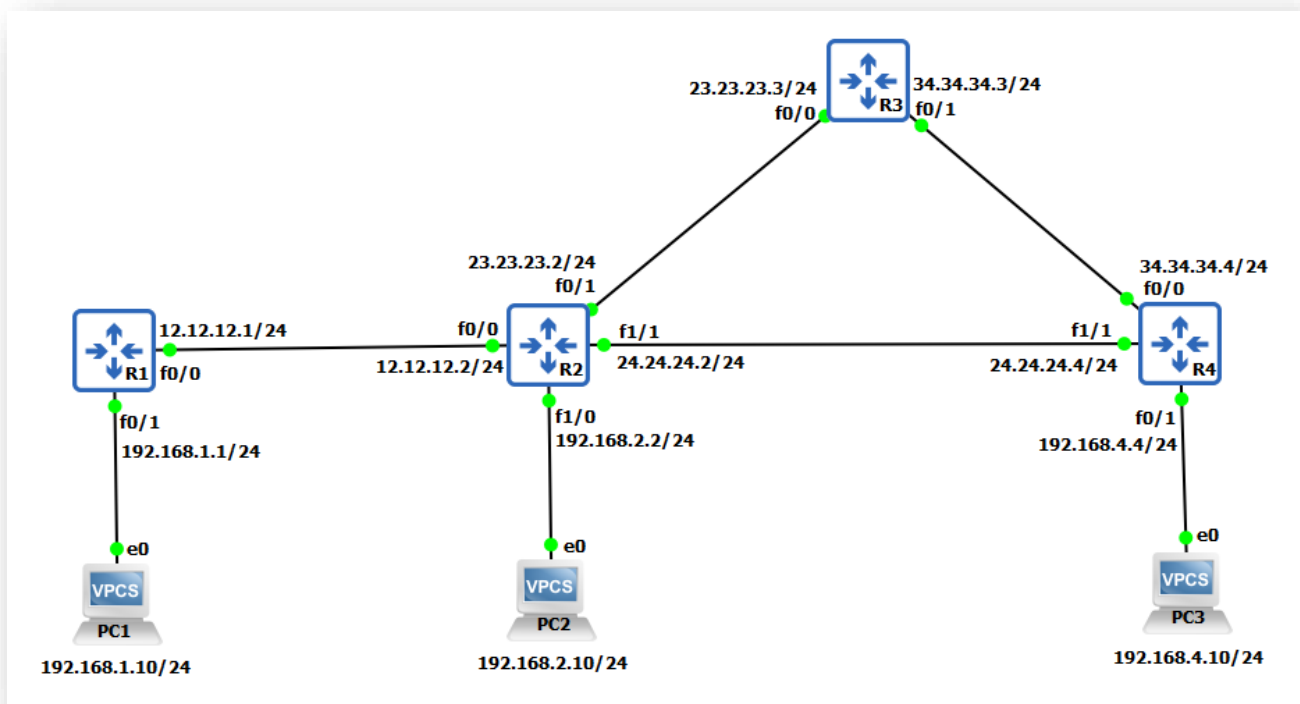


EIGRP



Task

1. Configure routers with IP address as shown in topology and configure enable password as **ccna** & device access password as **cisco (Refer Lab-1)**
2. Configure EIGRP process with ASN 100 on all routers and advertise its directly connected networks.
3. All router should able to ping each other's PCs IP.
4. PCs should able to ping each other's IP.
5. Modify default Hello and Hold-time on R1
6. Configure un-equal load balancing on R2 for R4's LAN network.
7. Configure Load as EIGRP metric for best path calculation.
8. Configure MD5 authentication between R2 & R4

Task-2: EIGRP Configuration with ASAN 100.

EIGRP Configuration on R1:

```

R1#config t
R1(config)#router eigrp 100
R1(config-router)#network 192.168.1.0
R1(config-router)#network 12.12.12.0
R1(config-router)#no auto-summary
R1(config-router)#exit
R1(config)#exit
R1#

```

EIGRP Configuration on R2:

```

R2#config t
R2(config)#router eigrp 100
R2(config-router)#network 12.12.12.0 255.255.255.0
R2(config-router)#network 23.23.23.0 255.255.255.0
R2(config-router)#network 24.24.24.0 255.255.255.0
R2(config-router)#network 192.168.2.0 255.255.255.0
R2(config-router)#no auto-summary
R2(config-router)#exit
R2(config)#exit
R2#

```

EIGRP Configuration on R3:

```

R3#config t
R3(config)#router eigrp 100
R3(config-router)#network 23.23.23.0 0.0.0.255
R3(config-router)#network 34.34.34.0 0.0.0.255
R3(config-router)#no auto-summary
R3(config-router)#exit
R3(config)#exit
R3#

```

EIGRP Configuration on R4:

```

R4#config t
R4(config)#router eigrp 100
R4(config-router)#network 24.24.24.4 0.0.0.0
R4(config-router)#network 34.34.34.4 0.0.0.0
R4(config-router)#network 192.168.4.0 0.0.0.255
R4(config-router)#no auto-summary
R4(config-router)#exit
R4(config)#exit
R4#

```

✓ Verification & Testing:**R1#sh ip protocols**

*** IP Routing is NSF aware ***

Routing Protocol is "eigrp 100"

Outgoing update filter list for all interfaces is not set
 Incoming update filter list for all interfaces is not set
 Default networks flagged in outgoing updates
 Default networks accepted from incoming updates

EIGRP-IPv4 Protocol for AS(100)

Metric weight K1=1, K2=0, K3=1, K4=0, K5=0

NSF-aware route hold timer is 240

Router-ID: 192.168.1.1

Topology : 0 (base)

Active Timer: 3 min

Distance: internal 90 external 170

Maximum path: 4

Maximum hopcount 100

Maximum metric variance 1

Automatic Summarization: disabled

Maximum path: 4

Routing for Networks:

12.0.0.0

192.168.1.0

Routing Information Sources:

Gateway	Distance	Last Update
12.12.12.2	90	01:05:28

12.12.12.2 90 01:05:28

Distance: internal 90 external 170

R2#sh ip protocols

*** IP Routing is NSF aware ***

Routing Protocol is "eigrp 100"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Default networks flagged in outgoing updates

Default networks accepted from incoming updates

EIGRP-IPv4 Protocol for AS(100)

Metric weight K1=1, K2=0, K3=1, K4=0, K5=0

NSF-aware route hold timer is 240

Router-ID: 192.168.2.2

Topology : 0 (base)

Active Timer: 3 min

Distance: internal 90 external 170

Maximum path: 4

Maximum hopcount 100

Maximum metric variance 1

Automatic Summarization: disabled

Maximum path: 4

Routing for Networks:

12.12.12.0/24

23.23.23.0/24

24.24.24.0/24

192.168.2.0

Routing Information Sources:

Gateway	Distance	Last Update
12.12.12.1	90	01:06:04
23.23.23.3	90	01:06:04
24.24.24.4	90	01:06:04

12.12.12.1 90 01:06:04

23.23.23.3 90 01:06:04

24.24.24.4 90 01:06:04

Distance: internal 90 external 170

R3#sh ip protocols

*** IP Routing is NSF aware ***

Routing Protocol is "eigrp 100"

Outgoing update filter list for all interfaces is not set
 Incoming update filter list for all interfaces is not set
 Default networks flagged in outgoing updates
 Default networks accepted from incoming updates

EIGRP-IPv4 Protocol for AS(100)

Metric weight K1=1, K2=0, K3=1, K4=0, K5=0

NSF-aware route hold timer is 240

Router-ID: 34.34.34.3

Topology : 0 (base)

Active Timer: 3 min

Distance: internal 90 external 170

Maximum path: 4

Maximum hopcount 100

Maximum metric variance 1

Automatic Summarization: disabled

Maximum path: 4

Routing for Networks:

23.23.23.0/24

34.34.34.0/24

Routing Information Sources:

Gateway	Distance	Last Update
23.23.23.2	90	01:06:11
34.34.34.4	90	01:06:11

23.23.23.2 90 01:06:11

34.34.34.4 90 01:06:11

Distance: internal 90 external 170

R4#sh ip protocols

*** IP Routing is NSF aware ***

Routing Protocol is "eigrp 100"

Outgoing update filter list for all interfaces is not set
 Incoming update filter list for all interfaces is not set
 Default networks flagged in outgoing updates
 Default networks accepted from incoming updates

EIGRP-IPv4 Protocol for AS(100)

Metric weight K1=1, K2=0, K3=1, K4=0, K5=0

NSF-aware route hold timer is 240

Router-ID: 192.168.4.4

Topology : 0 (base)

Active Timer: 3 min

Distance: internal 90 external 170

Maximum path: 4

Maximum hopcount 100

Maximum metric variance 1

Automatic Summarization: disabled

Maximum path: 4

Routing for Networks:

24.24.24.4/32

34.34.34.4/32

192.168.4.0

Routing Information Sources:

Gateway	Distance	Last Update
24.24.24.2	90	01:06:15
34.34.34.3	90	01:12:16

Distance: internal 90 external 170

R1#sh ip eigrp neighbors

EIGRP-IPv4 Neighbors for AS(100)

H	Address	Interface	Hold Uptime (sec)	SRTT (ms)	RTO	Q Cnt	Seq Num
0	12.12.12.2	Fa0/0	13 01:14:37	19	114	0	25

R1#

R2#sh ip eigrp neighbors

EIGRP-IPv4 Neighbors for AS(100)

H	Address	Interface	Hold Uptime (sec)	SRTT (ms)	RTO	Q Cnt	Seq Num
2	24.24.24.4	Fa1/1	13 01:14:27	21	126	0	7
1	23.23.23.3	Fa0/1	14 01:14:37	24	144	0	10
0	12.12.12.1	Fa0/0	12 01:14:49	24	144	0	8

R2#

R3#sh ip eigrp neighbors

EIGRP-IPv4 Neighbors for AS(100)

H	Address	Interface	Hold Uptime (sec)	SRTT (ms)	RTO	Q Cnt	Seq Num
1	34.34.34.4	Fa0/1	12 01:14:30	14	100	0	6
0	23.23.23.2	Fa0/0	14 01:14:40	668	4008	0	22

R3#

R4#sh ip eigrp neighbors

EIGRP-IPv4 Neighbors for AS(100)

H	Address	Interface	Hold Uptime (sec)	SRTT (ms)	RTO	Q Cnt	Seq Num
1	34.34.34.3	Fa0/0	11 01:14:32	1586	5000	0	11
0	24.24.24.2	Fa1/1	13 01:14:32	1244	5000	0	23

R4#

R1#sh ip route eigrp

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 + - replicated route, % - next hop override

Gateway of last resort is not set

23.0.0.0/24 is subnetted, 1 subnets

D 23.23.23.0 [90/30720] via 12.12.12.2, 00:01:45, FastEthernet0/0

24.0.0.0/24 is subnetted, 1 subnets

D 24.24.24.0 [90/30720] via 12.12.12.2, 00:01:45, FastEthernet0/0

34.0.0.0/24 is subnetted, 1 subnets

D 34.34.34.0 [90/33280] via 12.12.12.2, 00:01:37, FastEthernet0/0

D 192.168.2.0/24 [90/30720] via 12.12.12.2, 00:01:45, FastEthernet0/0

D 192.168.4.0/24 [90/33280] via 12.12.12.2, 00:01:28, FastEthernet0/0

R1#

R2#sh ip route eigrp | begin Gateway

Gateway of last resort is not set

34.0.0.0/24 is subnetted, 1 subnets

D 34.34.34.0 [90/30720] via 24.24.24.4, 00:02:01, FastEthernet1/1

[90/30720] via 23.23.23.3, 00:02:01, FastEthernet0/1

D 192.168.1.0/24 [90/30720] via 12.12.12.1, 00:02:18, FastEthernet0/0

D 192.168.4.0/24 [90/30720] via 24.24.24.4, 00:02:01, FastEthernet1/1

R2#

R3#sh ip route eigrp | begin Gateway

Gateway of last resort is not set

12.0.0.0/24 is subnetted, 1 subnets

D 12.12.12.0 [90/30720] via 23.23.23.2, 00:02:21, FastEthernet0/0

24.0.0.0/24 is subnetted, 1 subnets

D 24.24.24.0 [90/30720] via 34.34.34.4, 00:02:26, FastEthernet0/1

[90/30720] via 23.23.23.2, 00:02:26, FastEthernet0/0

D 192.168.1.0/24 [90/33280] via 23.23.23.2, 00:02:21, FastEthernet0/0

D 192.168.2.0/24 [90/30720] via 23.23.23.2, 00:02:21, FastEthernet0/0

D 192.168.4.0/24 [90/30720] via 34.34.34.4, 00:02:26, FastEthernet0/1

R3#

R4#sh ip route eigrp | begin Gateway

Gateway of last resort is not set

12.0.0.0/24 is subnetted, 1 subnets

D 12.12.12.0 [90/30720] via 24.24.24.2, 00:02:28, FastEthernet1/1

23.0.0.0/24 is subnetted, 1 subnets

D 23.23.23.0 [90/30720] via 34.34.34.3, 00:02:28, FastEthernet0/0

[90/30720] via 24.24.24.2, 00:02:28, FastEthernet1/1

D 192.168.1.0/24 [90/33280] via 24.24.24.2, 00:02:28, FastEthernet1/1

D 192.168.2.0/24 [90/30720] via 24.24.24.2, 00:02:28, FastEthernet1/1

R4#

R1#sh ip eigrp topology

EIGRP-IPv4 Topology Table for AS(100)/ID(192.168.1.1)

Codes: **P** - Passive, A - Active, U - Update, Q - Query, R - Reply,

r - reply Status, s - sia Status

P 34.34.34.0/24, 1 successors, FD is 33280

via 12.12.12.2 (33280/30720), FastEthernet0/0
 P 192.168.2.0/24, 1 successors, FD is 30720
 via 12.12.12.2 (30720/28160), FastEthernet0/0
 P 23.23.23.0/24, 1 successors, FD is 30720
 via 12.12.12.2 (30720/28160), FastEthernet0/0
 P 12.12.12.0/24, 1 successors, FD is 28160
 via Connected, FastEthernet0/0
 P 24.24.24.0/24, 1 successors, FD is 30720
 via 12.12.12.2 (30720/28160), FastEthernet0/0
 P 192.168.1.0/24, 1 successors, FD is 28160
 via Connected, FastEthernet0/1
 P 192.168.4.0/24, 1 successors, FD is 33280
 via 12.12.12.2 (33280/30720), FastEthernet0/0

R2#sh ip eigrp topology

EIGRP-IPv4 Topology Table for AS(100)/ID(192.168.2.2)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
 r - reply Status, s - sia Status

P 34.34.34.0/24, 2 successors, FD is 30720
 via 23.23.23.3 (30720/28160), FastEthernet0/1
 via 24.24.24.4 (30720/28160), FastEthernet1/1
 P 192.168.2.0/24, 1 successors, FD is 28160
 via Connected, FastEthernet1/0
 P 23.23.23.0/24, 1 successors, FD is 28160
 via Connected, FastEthernet0/1
 P 12.12.12.0/24, 1 successors, FD is 28160
 via Connected, FastEthernet0/0
 P 24.24.24.0/24, 1 successors, FD is 28160
 via Connected, FastEthernet1/1
 P 192.168.1.0/24, 1 successors, FD is 30720
 via 12.12.12.1 (30720/28160), FastEthernet0/0
 P 192.168.4.0/24, 1 successors, FD is 30720
 via 24.24.24.4 (30720/28160), FastEthernet1/1

R3#sh ip eigrp topology

EIGRP-IPv4 Topology Table for AS(100)/ID(34.34.34.3)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
 r - reply Status, s - sia Status

P 34.34.34.0/24, 1 successors, FD is 28160
 via Connected, FastEthernet0/1
 P 192.168.2.0/24, 1 successors, FD is 30720
 via 23.23.23.2 (30720/28160), FastEthernet0/0
 P 23.23.23.0/24, 1 successors, FD is 28160
 via Connected, FastEthernet0/0
 P 12.12.12.0/24, 1 successors, FD is 30720
 via 23.23.23.2 (30720/28160), FastEthernet0/0
 P 24.24.24.0/24, 2 successors, FD is 30720
 via 23.23.23.2 (30720/28160), FastEthernet0/0
 via 34.34.34.4 (30720/28160), FastEthernet0/1

P 192.168.1.0/24, 1 successors, FD is 33280
via 23.23.23.2 (33280/30720), FastEthernet0/0
P 192.168.4.0/24, 1 successors, FD is 30720
via 34.34.34.4 (30720/28160), FastEthernet0/1

R4#sh ip eigrp topology

EIGRP-IPv4 Topology Table for AS(100)/ID(192.168.4.4)
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
r - reply Status, s - sia Status

P 34.34.34.0/24, 1 successors, FD is 28160
via Connected, FastEthernet0/0
P 192.168.2.0/24, 1 successors, FD is 30720
via 24.24.24.2 (30720/28160), FastEthernet1/1
P 23.23.23.0/24, 2 successors, FD is 30720
via 24.24.24.2 (30720/28160), FastEthernet1/1
via 34.34.34.3 (30720/28160), FastEthernet0/0
P 12.12.12.0/24, 1 successors, FD is 30720
via 24.24.24.2 (30720/28160), FastEthernet1/1
P 24.24.24.0/24, 1 successors, FD is 28160
via Connected, FastEthernet1/1
P 192.168.1.0/24, 1 successors, FD is 33280
via 24.24.24.2 (33280/30720), FastEthernet1/1
P 192.168.4.0/24, 1 successors, FD is 28160
via Connected, FastEthernet0/1

R1#sh ip eigrp traffic

EIGRP-IPv4 Traffic Statistics for AS(100)
Hellos sent/received: 7167/4648
Updates sent/received: 13/18
Queries sent/received: 0/1
Replies sent/received: 1/0
Acks sent/received: 17/11
SIA-Queries sent/received: 0/0
SIA-Replies sent/received: 0/0
Hello Process ID: 209
PDM Process ID: 195
Socket Queue: 0/10000/2/0 (current/max/highest/drops)
Input Queue: 0/2000/2/0 (current/max/highest/drops)

R2#sh ip eigrp traffic

EIGRP-IPv4 Traffic Statistics for AS(100)
Hellos sent/received: 18637/11826
Updates sent/received: 48/37
Queries sent/received: 4/1
Replies sent/received: 1/4
Acks sent/received: 31/47
SIA-Queries sent/received: 0/0
SIA-Replies sent/received: 0/0
Hello Process ID: 209
PDM Process ID: 195

Socket Queue: 0/10000/4/0 (current/max/highest/drops)
Input Queue: 0/2000/4/0 (current/max/highest/drops)

R3#sh ip eigrp traffic

EIGRP-IPv4 Traffic Statistics for AS(100)

Hellos sent/received: 9325/9317

Updates sent/received: 25/29

Queries sent/received: 3/5

Replies sent/received: 5/3

Acks sent/received: 35/24

SIA-Queries sent/received: 0/0

SIA-Replies sent/received: 0/0

Hello Process ID: 209

PDM Process ID: 195

Socket Queue: 0/10000/2/0 (current/max/highest/drops)

Input Queue: 0/2000/2/0 (current/max/highest/drops)

R4#sh ip eigrp traffic

EIGRP-IPv4 Traffic Statistics for AS(100)

Hellos sent/received: 13978/9314

Updates sent/received: 27/25

Queries sent/received: 3/3

Replies sent/received: 3/3

Acks sent/received: 25/26

SIA-Queries sent/received: 0/0

SIA-Replies sent/received: 0/0

Hello Process ID: 209

PDM Process ID: 195

Socket Queue: 0/10000/2/0 (current/max/highest/drops)

Input Queue: 0/2000/2/0 (current/max/highest/drops)

Ping PC3 and PC2 from R1**R1#ping 192.168.4.10**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.4.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 48/55/76 ms

R1#ping 192.168.2.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.2.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 20/28/40 ms

R1#

Ping PC1 and PC3 from R2**R2#ping 192.168.1.10**

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 20/28/40 ms

R2#

R2#ping 192.168.4.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.4.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 12/24/28 ms

R2#

Ping PC1, PC2 and PC3 from R3

R3#ping 192.168.1.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/48/52 ms

R3#

R3#ping 192.168.2.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.2.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 4/24/32 ms

R3#

R3#ping 192.168.4.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.4.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 12/24/28 ms

R3#

Ping PC1 and PC2 from R3

R4#ping 192.168.1.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 48/50/52 ms

R4#

R4#ping 192.168.2.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.2.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 16/25/28 ms

R4#

Ping PC2 & PC3 from PC1**PC1> ping 192.168.2.10**

192.168.2.10 icmp_seq=1 timeout

192.168.2.10 icmp_seq=2 timeout

84 bytes from 192.168.2.10 icmp_seq=3 ttl=62 time=34.187 ms

84 bytes from 192.168.2.10 icmp_seq=4 ttl=62 time=36.313 ms

84 bytes from 192.168.2.10 icmp_seq=5 ttl=62 time=38.284 ms

PC1> ping 192.168.4.10

192.168.4.10 icmp_seq=1 timeout

192.168.4.10 icmp_seq=2 timeout

84 bytes from 192.168.4.10 icmp_seq=3 ttl=61 time=65.192 ms

84 bytes from 192.168.4.10 icmp_seq=4 ttl=61 time=57.046 ms

84 bytes from 192.168.4.10 icmp_seq=5 ttl=61 time=63.101 ms

Ping PC1 & PC3 from PC2**PC2> ping 192.168.1.10**

84 bytes from 192.168.1.10 icmp_seq=1 ttl=62 time=38.294 ms

84 bytes from 192.168.1.10 icmp_seq=2 ttl=62 time=32.358 ms

84 bytes from 192.168.1.10 icmp_seq=3 ttl=62 time=40.143 ms

84 bytes from 192.168.1.10 icmp_seq=4 ttl=62 time=37.042 ms

84 bytes from 192.168.1.10 icmp_seq=5 ttl=62 time=43.391 ms

PC2> ping 192.168.4.10

84 bytes from 192.168.4.10 icmp_seq=1 ttl=62 time=36.284 ms

84 bytes from 192.168.4.10 icmp_seq=2 ttl=62 time=39.322 ms

84 bytes from 192.168.4.10 icmp_seq=3 ttl=62 time=40.337 ms

84 bytes from 192.168.4.10 icmp_seq=4 ttl=62 time=44.225 ms

84 bytes from 192.168.4.10 icmp_seq=5 ttl=62 time=42.331 ms

Ping PC1 & PC2 from PC3**PC3> ping 192.168.1.10**

84 bytes from 192.168.1.10 icmp_seq=1 ttl=61 time=54.332 ms

84 bytes from 192.168.1.10 icmp_seq=2 ttl=61 time=66.209 ms

84 bytes from 192.168.1.10 icmp_seq=3 ttl=61 time=58.276 ms

84 bytes from 192.168.1.10 icmp_seq=4 ttl=61 time=64.044 ms

84 bytes from 192.168.1.10 icmp_seq=5 ttl=61 time=62.304 ms

PC3> ping 192.168.2.10

84 bytes from 192.168.2.10 icmp_seq=1 ttl=62 time=37.324 ms

84 bytes from 192.168.2.10 icmp_seq=2 ttl=62 time=36.298 ms

84 bytes from 192.168.2.10 icmp_seq=3 ttl=62 time=41.267 ms

84 bytes from 192.168.2.10 icmp_seq=4 ttl=62 time=38.175 ms

84 bytes from 192.168.2.10 icmp_seq=5 ttl=62 time=37.332 ms

Ping EIGRP multicast IP from all routers**R1#ping 224.0.0.10**

Type escape sequence to abort.

Sending 1, 100-byte ICMP Echos to 224.0.0.10, timeout is 2 seconds:

Reply to request 0 from 12.12.12.2, 8 ms
R1#

R2#ping 224.0.0.10

Type escape sequence to abort.

Sending 1, 100-byte ICMP Echos to 224.0.0.10, timeout is 2 seconds:

Reply to request 0 from 23.23.23.3, 4 ms
Reply to request 0 from 24.24.24.4, 8 ms
Reply to request 0 from 12.12.12.1, 4 ms
R2#

R3#ping 224.0.0.10

Type escape sequence to abort.

Sending 1, 100-byte ICMP Echos to 224.0.0.10, timeout is 2 seconds:

Reply to request 0 from 34.34.34.4, 12 ms
Reply to request 0 from 23.23.23.2, 12 ms
R3#

R4#ping 224.0.0.10

Type escape sequence to abort.

Sending 1, 100-byte ICMP Echos to 224.0.0.10, timeout is 2 seconds:

Reply to request 0 from 34.34.34.3, 8 ms
Reply to request 0 from 24.24.24.2, 12 ms
R4#

Task-5: Configure Hello interval of 10 sec and hold-time of 30 sec on R1

R1 Configuration:

```
R1#config t
R1(config)#int fa0/0
R1(config-if)#ip hello-interval eigrp 100 10
R1(config-if)#ip hold-time eigrp 100 30
R1(config-if)#exit
R1(config)#exit
R1#
```

✓ Verification & Testing:

R1#sh ip eigrp interfaces detail fa0/0

EIGRP-IPv4 Interfaces for AS(100)

Interface	Peers	Xmit Queue Un/Reliable	PeerQ Un/Reliable	Mean SRTT	Pacing Time Un/Reliable	Multicast Flow Timer	Pending Routes
Fa0/0	1	0/0	0/0	12	0/0	50	0

Hello-interval is 10, Hold-time is 30

Split-horizon is enabled
 Next xmit serial <none>
 Packetized sent/expedited: 3/0
 Hello's sent/expedited: 539/2
 Un/reliable mcasts: 0/4 Un/reliable ucasts: 4/2
 Mcast exceptions: 0 CR packets: 0 ACKs suppressed: 0
 Retransmissions sent: 1 Out-of-sequence rcvd: 1
 Topology-ids on interface - 0
 Authentication mode is not set

R1#

R2#sh ip eigrp interfaces detail fa0/0

EIGRP-IPv4 Interfaces for AS(100)

Interface	Peers	Xmit Queue Un/Reliable	PeerQ Un/Reliable	Mean SRTT	Pacing Time Un/Reliable	Multicast Flow Timer	Pending Routes
Fa0/0	1	0/0	0/0	1032	0/0	5140	0

Hello-interval is 5, Hold-time is 15

Split-horizon is enabled
 Next xmit serial <none>
 Packetized sent/expedited: 4/0
 Hello's sent/expedited: 729/2
 Un/reliable mcasts: 0/4 Un/reliable ucasts: 5/2
 Mcast exceptions: 0 CR packets: 0 ACKs suppressed: 0
 Retransmissions sent: 1 Out-of-sequence rcvd: 0
 Topology-ids on interface - 0
 Authentication mode is not set

R2#

R1#sh ip eigrp neighbors

EIGRP-IPv4 Neighbors for AS(100)

H	Address	Interface	Hold Uptime (sec)	SRTT (ms)	RTO	Q Cnt	Seq Num
0	12.12.12.2	Fa0/0	12 01:13:33	12	100	0	12

R1#

R2#sh ip eigrp neighbors

EIGRP-IPv4 Neighbors for AS(100)

H	Address	Interface	Hold Uptime (sec)	SRTT (ms)	RTO	Q Cnt	Seq Num
2	24.24.24.4	Fa1/1	11 01:13:48	10	100	0	6
1	23.23.23.3	Fa0/1	12 01:13:58	68	408	0	7
0	12.12.12.1	Fa0/0	29 01:14:05	1032	5000	0	5

R2#

Note: It is recommended to configure EIGRP Hello and Hold-time timers on both ends. If you configure the timers only on one end, a unidirectional link occurs and there can be issue with neighborship.

Task 6: Configure un-equal load balancing on R2 for R4's LAN network.**R2#sh ip eigrp topology all-links**

EIGRP-IPv4 Topology Table for AS(100)/ID(192.168.2.2)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
r - reply Status, s - sia Status

```

P 34.34.34.0/24, 2 successors, FD is 30720, serno 7
  via 23.23.23.3 (30720/28160), FastEthernet0/1
  via 24.24.24.4 (30720/28160), FastEthernet1/1
P 192.168.2.0/24, 1 successors, FD is 28160, serno 4
  via Connected, FastEthernet1/0
P 23.23.23.0/24, 1 successors, FD is 28160, serno 2
  via Connected, FastEthernet0/1
P 12.12.12.0/24, 1 successors, FD is 28160, serno 1
  via Connected, FastEthernet0/0
P 24.24.24.0/24, 1 successors, FD is 5120, serno 9
  via Connected, FastEthernet1/1
P 192.168.1.0/24, 1 successors, FD is 30720, serno 5
  via 12.12.12.1 (30720/28160), FastEthernet0/0
P 192.168.4.0/24, 1 successors, FD is 30720, serno 8
  via 24.24.24.4 (30720/28160), FastEthernet1/1
  via 23.23.23.3 (33280/30720), FastEthernet0/1

```

Note: EIGRP provides a mechanism to load balance over unequal cost paths through Variance Command. Variance is a number (1 to 128), multiplied by the local best metric then includes the routes with the lesser or equal metric. The default Variance value is 1, which means equal-cost load balancing.

Variance configuration on R2

```

R2#config t
R2(config)#router eigrp 100
R2(config-router)#v
R2(config-router)#variance 2
R2(config-router)#exit
R2(config)#exit
R2#

```

✓ Verification & Testing:**R2#sh ip protocols**

*** IP Routing is NSF aware ***

Routing Protocol is "eigrp 100"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Default networks flagged in outgoing updates

Default networks accepted from incoming updates

EIGRP-IPv4 Protocol for AS(100)

Metric weight K1=1, K2=0, K3=1, K4=0, K5=0

NSF-aware route hold timer is 240

Router-ID: 192.168.2.2

Topology : 0 (base)

Active Timer: 3 min

Distance: internal 90 external 170

Maximum path: 4

Maximum hopcount 100

Maximum metric variance 2

Automatic Summarization: disabled

Maximum path: 4

Routing for Networks:

12.12.12.0/24

23.23.23.0/24

24.24.24.0/24

192.168.2.0

Routing Information Sources:

Gateway	Distance	Last Update
12.12.12.1	90	01:06:04
23.23.23.3	90	01:06:04
24.24.24.4	90	01:06:04

Distance: internal 90 external 170

R2#

R2#sh ip route eigrp

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

+ - replicated route, % - next hop override

Gateway of last resort is not set

34.0.0.0/24 is subnetted, 1 subnets

D 34.34.34.0 [90/30720] via 24.24.24.4, 00:02:36, FastEthernet1/1

[90/30720] via 23.23.23.3, 00:02:36, FastEthernet0/1

D 192.168.1.0/24 [90/30720] via 12.12.12.1, 00:02:36, FastEthernet0/0

D 192.168.4.0/24 [90/30720] via 24.24.24.4, 00:02:36, FastEthernet1/1

[90/33280] via 23.23.23.3, 00:02:36, FastEthernet0/1

R2#

R4#sh ip route eigrp | begin Gateway

Gateway of last resort is not set

12.0.0.0/24 is subnetted, 1 subnets

D 12.12.12.0 [90/30720] via 24.24.24.2, 02:47:19, FastEthernet1/1

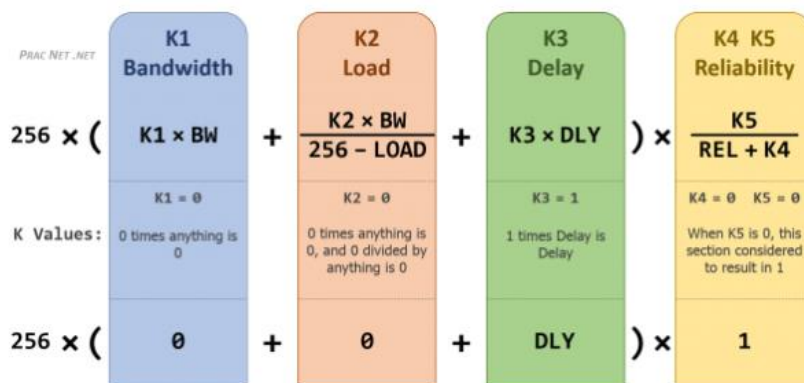
23.0.0.0/24 is subnetted, 1 subnets

D 23.23.23.0 [90/30720] via 34.34.34.3, 02:47:19, FastEthernet0/0

[90/30720] via 24.24.24.2, 02:47:19, FastEthernet1/1
D 192.168.1.0/24 [90/33280] via 24.24.24.2, 02:47:19, FastEthernet1/1
D 192.168.2.0/24 [90/30720] via 24.24.24.2, 02:47:19, FastEthernet1/1
R4#

Task 7: Configure Load as EIGRP metric for best path calculation on all routers.

EIGRP Metric Calculation



R1 Configuration:

```
R1#config t
R1(config)#router eigrp 100
R1(config-router)#metr
R1(config-router)#metric we
R1(config-router)#metric weights ?
<0-8> Type (Only TOS 0 supported)
R1(config-router)#metric weights 0 ?
<0-255> K1
R1(config-router)#metric weights 0 0 ?
<0-255> K2
R1(config-router)#metric weights 0 0 0 ?
<0-255> K3
R1(config-router)#metric weights 0 0 0 1 ?
<0-255> K4
R1(config-router)#metric weights 0 0 0 1 0 ?
<0-255> K5
R1(config-router)#metric weights 0 0 0 1 0 0
R1(config-router)#exit
R1(config)#exit
R1#
```

*Oct 23 19:38:00.683: %DUAL-5-NBRCHANGE: EIGRP-IPv4 100: Neighbor 12.12.12.2 (FastEthernet0/0) is down: K-value mismatch

R2 Configuration:

```
R2#config t
R2(config)#router eigrp 100
R2(config-router)#metric weights 0 0 0 1 0 0
R2(config-router)#exit
R2(config)#exit
R2#
```

```
*Oct 23 19:39:12.739: %DUAL-5-NBRCHANGE: EIGRP-IPv4 100: Neighbor 24.24.24.4
(FastEthernet1/1) is down: metric changed
*Oct 23 19:39:12.743: %DUAL-5-NBRCHANGE: EIGRP-IPv4 100: Neighbor 23.23.23.3
(FastEthernet0/1) is down: metric changed
*Oct 23 19:39:13.227: %DUAL-5-NBRCHANGE: EIGRP-IPv4 100: Neighbor 23.23.23.3
(FastEthernet0/1) is down: K-value mismatch
```

R3 Configuration:

```
R3#config t
R3(config)#router eigrp 100
R3(config-router)#metric weights 0 0 0 1 0 0
R3(config-router)#exit
R3(config)#exit
R3#
```

```
*Oct 23 19:42:45.347: %DUAL-5-NBRCHANGE: EIGRP-IPv4 100: Neighbor 34.34.34.4
(FastEthernet0/1) is down: metric changed
```

R4 Configuration:

```
R4#config t
R4(config)#router eigrp 100
R4(config-router)#metric weights 0 0 0 1 0 0
R4(config-router)#exit
R4(config)#exit
R4#
```

```
*Oct 23 19:42:51.447: %DUAL-5-NBRCHANGE: EIGRP-IPv4 100: Neighbor 24.24.24.2
(FastEthernet1/1) is up: new adjacency
*Oct 23 19:42:52.127: %DUAL-5-NBRCHANGE: EIGRP-IPv4 100: Neighbor 34.34.34.3
(FastEthernet0/0) is up: new adjacency
```

✓ Verification & Testing:**R1#sh ip protocols | section Metric**

```
Metric weight K1=0, K2=0, K3=1, K4=0, K5=0
R1#
```

R2#sh ip protocols | section Metric

```
Metric weight K1=0, K2=0, K3=1, K4=0, K5=0
R2#
```

R3#sh ip protocols | section Metric

Metric weight K1=0, K2=0, K3=1, K4=0, K5=0

R3#

R4#sh ip protocols | section Metric

Metric weight K1=0, K2=0, K3=1, K4=0, K5=0

R4#

Task 8: Configure MD5 authentication on R2 & R4 with key string Cisco**R2 Configuration:**

R2#config t

R2(config)#key chain **MyChain**

R2(config-keychain)#key 1

R2(config-keychain-key)#key-string Cisco

R2(config-keychain-key)#exit

R2(config-keychain)#int fa1/1

R2(config-if)#ip authentication key-chain eigrp 100 MyChain

R2(config-if)#ip authentication mode eigrp 100 md5

R2(config-if)#exit

R2(config)#exit

R2#

*Oct 23 20:05:06.479: %DUAL-5-NBRCHANGE: EIGRP-IPv4 100: Neighbor 24.24.24.4 (FastEthernet1/1) is down: authentication mode changed

R4 Configuration:

R4#config t

R4(config)#key chain **YourChain**

R4(config-keychain)#key 1

R4(config-keychain-key)#key-string Cisco

R4(config-keychain-key)#exit

R4(config-keychain)#int fa1/1

R4(config-if)#ip authentication key-chain eigrp 100 YourChain

R4(config-if)#ip authentication mode eigrp 100 md5

R4(config-if)#exit

R4(config)#exit

R4#

*Oct 23 20:05:35.303: %SYS-5-CONFIG_I: Configured from console by console

*Oct 23 20:05:37.459: %DUAL-5-NBRCHANGE: EIGRP-IPv4 100: Neighbor 24.24.24.2 (FastEthernet1/1) is up: new adjacency

✓ Verification & Testing:**R2#sh ip eigrp interfaces detail fa1/1**

EIGRP-IPv4 Interfaces for AS(100)

Interface	Peers	Xmit Queue Un/Reliable	PeerQ Un/Reliable	Mean SRTT	Pacing Time Un/Reliable	Multicast Flow Timer	Pending Routes
Fa1/1	1	0/0	0/0	1597	0/1	7968	0

Hello-interval is 5, Hold-time is 15
Split-horizon is enabled
Next xmit serial <none>
Packetized sent/expedited: 11/0
Hello's sent/expedited: 2767/4
Un/reliable mcasts: 0/10 Un/reliable ucasts: 9/5
Mcast exceptions: 0 CR packets: 0 ACKs suppressed: 0
Retransmissions sent: 2 Out-of-sequence rcvd: 3
Topology-ids on interface - 0
Authentication mode is md5, key-chain is "MyChain"

R2#

R4#sh ip eigrp interfaces detail fa1/1 | section Authentication

Authentication mode is md5, key-chain is "YourChain"

R4#

R2#sh key chain

Key-chain MyChain:

key 1 -- text "Cisco"
accept lifetime (always valid) - (always valid) [valid now]
send lifetime (always valid) - (always valid) [valid now]

R4#sh key chain

Key-chain YourChain:

key 1 -- text "Cisco"
accept lifetime (always valid) - (always valid) [valid now]
send lifetime (always valid) - (always valid) [valid now]

R4#

Important Commands:

```
sh ip protocols
sh ip eigrp neighbors
sh ip route eigrp
sh ip route eigrp | begin Gateway
sh ip eigrp topology
sh ip eigrp traffic
sh ip eigrp interfaces detail fax/x
sh ip eigrp topology all-links
sh ip protocols | section Metric
sh ip eigrp interfaces detail fax/x | section Authentication
sh key chain
```

Ratan